

LAB 1

NETWORK CABLING: PREPARATION OF NETWORK CABLES AND THEIR USES

OBJECTIVES :

- Learn the fundamentals of Ethernet cabling used in computer networks.
- Identify end devices (Category A) and networking devices (Category B) based on their transmit (TX) and receive (RX) pins.
- Understand the principles of network cable termination and connectivity.
- Develop practical skills in testing and troubleshooting Ethernet cables for reliable network performance.

COMPONENTS REQUIRED :

- i. Twisted Pair Cable (Cat 5e or 6)
- ii. RJ45 connector
- iii. Crimping tool (Clamper)
- iv. Network Cable Tester

THEORY :

Ethernet is the backbone of most Local Area Networks (LANs). It offers a standardized way to connect computers and networking devices using twisted-pair cables such as Cat5 and Cat6. An Ethernet cable is built from eight copper wires grouped into four twisted pairs, all ending in an RJ-45 connector. To ensure proper communication, each wire follows a color code defined by standards like TIA/EIA-568A or TIA/EIA-568B. Sticking to these codes is essential for error-free data transmission.

When it comes to communication, Ethernet devices fall into two categories based on how their pins are assigned:

- **Category A – End Devices:** PCs, laptops, and printers. These send data on pins 1 and 2 (TX) and receive on pins 3 and 6 (RX).
- **Category B – Networking Devices:** Switches and hubs. These send data on pins 3 and 6 (TX) and receive on pins 1 and 2 (RX).

CONNECTION TYPE :

1. Straight through cables:

Straight-through cables are used to connect devices from different categories—typically an end device (Category A), such as a computer, laptop, or printer, to a networking device (Category B), like a switch or router. In this configuration, both ends of the cable follow the same wiring standard, either T568A–T568A or T568B–T568B. This ensures that the transmit

(TX) pins on the end device align with the receive (RX) pins on the networking device, allowing proper communication. Straight-through cables are the most common type used in LAN setups.

In the lab, we followed T568A-T568A standard and the required connection was done as shown below :

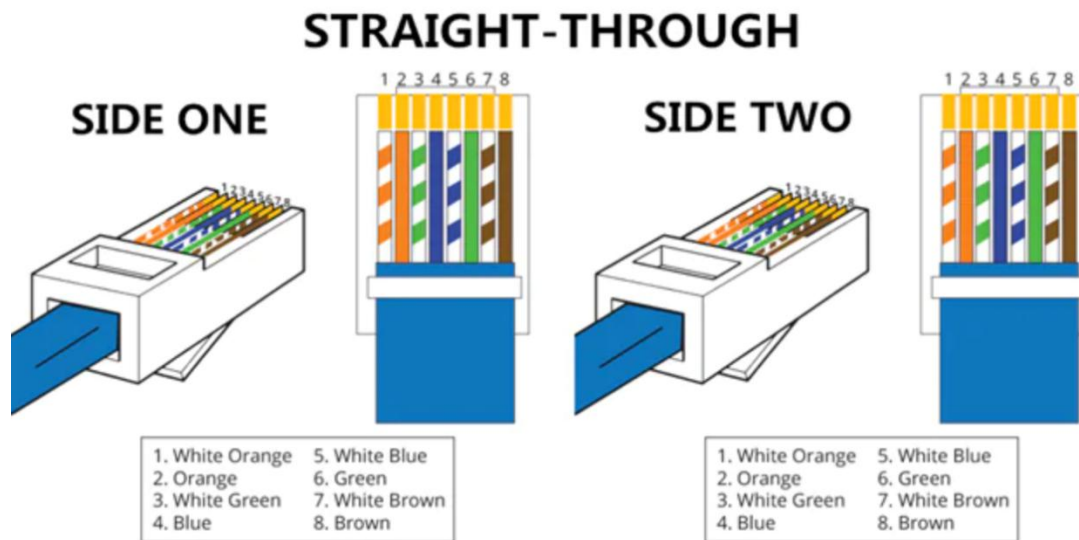


Fig: Straight Through Cabling Connection

2. Crossover Cabling:

Crossover cables, on the other hand, are designed to connect devices of the same category—either two end devices (Category A ↔ Category A) or two networking devices (Category B ↔ Category B). One end of the cable is wired using the T568A standard, while the other end uses T568B. This crossover in wiring swaps the TX and RX pins, enabling direct communication between similar devices without the need for an intermediary switch or hub. Crossover cables are especially useful in peer-to-peer setups or for testing network hardware.

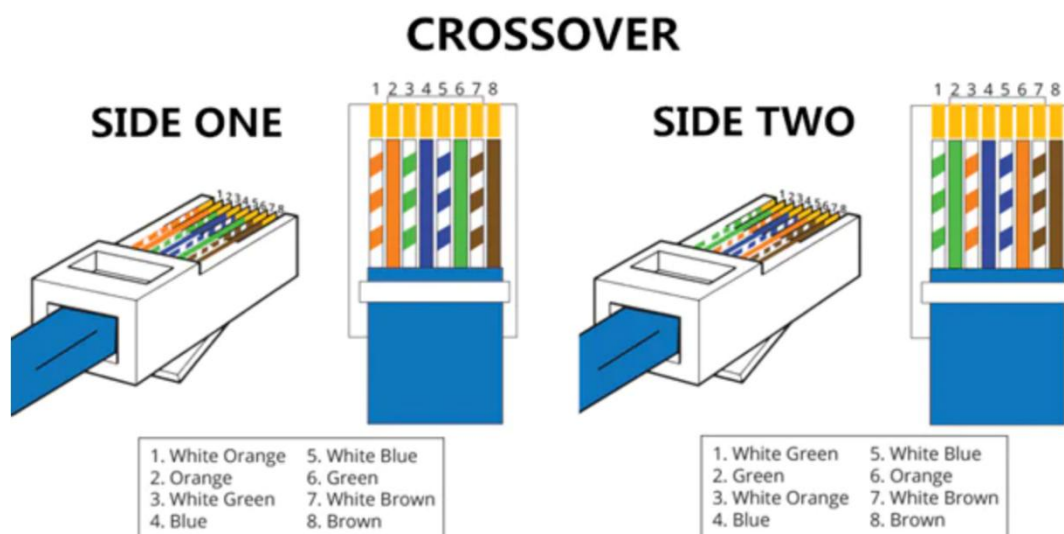


Fig: Crossover Cabling Connection

Straight-Through vs Crossover Cable Comparison

Feature	Straight-Through Cable	Crossover Cable
Wiring Standard	Both ends same (T568A–T568A or T568B–T568B)	Different ends (T568A–T568B)
Device Connection Type	Different devices (PC → Switch/Router)	Similar devices (PC → PC, Switch → Switch)
Signal Direction	Transmit/Receive straight	Transmit/Receive crossed
Common Use	Daily network connections	Direct testing or peer-to-peer link

OBSERVATION:

The observation was done on the Network LAN tester for RJ45 with both straight-through and crossover. The connection was shown by the blinking led which is numbered on the figure shown below :



Fig: Network LAN Tester

DISCUSSION

In the lab, we understood about the preparation of network cables with their applications where straight-through cabling is used for connecting devices which fall on different category and crossover through cabling is used for connecting devices which fall on same category. On our first try we did wrong connection by performing connection from left to right on one end and on the other end by going from right to left direction for crossover cabling. With the proper guidance of our lab teacher, we were able to make another one within the allocated time.

CONCLUSION

Thus, network cabling (i.e. the preparation of network cable) was carried out in the lab through a combination of theoretical understanding and practical exposure, enabling us to gain valuable hands-on experience. By doing the work ourselves, we learned the importance of following correct wiring standards and using the right cable for each situation.